# Finding inverse modulo a prime

Pushpendra Pal

## An example of finding inverse in a prime field

We take 357 and 95, GCD(357,95) = 1, 357 is not prime, but we can still find the value such that 357x + 95y = 1.

### GCD Euclidean

$R_{-1}$ = 357, $R_0$ = 95

| index | $R_{i-2}$ | = | $R_{i-1}$ | * $F_i$ | + $R_i$ |
|---|---|---|---|---|---|
| 1 | 357 | = | 95 | *(3) | + 72 |
| 2 | 95 | = | 72 | *(1) | + 23 |
| 3 | 72 | = | 23 | *(3) | + 3 |
| 4 | 23 | = | 3 | *(7) | + 2 |
| 5 | 3 | = | 2 | *(1) | + 1 |

### Inverse

| $R_i$ = | $R_{i-2}$ | * $COEFF_{i-2}$ | + $R_{i-1}$ | * $COEFF_{i-1}$ |
|---|---|---|---|---|
| $R_5$ = | $R_3$ | * (1) | + $R_4$ | * $(-F_5)$ |
| $R_5$ = | $R_2$ | * $(-F_5)$ | + $R_3$ | * $(1+F_4F_5)$ |
| $R_5$ = | $R_1$ | * $(1+F_4F_5)$ | + $R_2$ | * $(-F_3-F_5-F_3F_4F_5)$ |
| $R_5$ = | $R_0$ | * $(-F_3-F_5-F_3F_4F_5)$ | + $R_1$ | * $(1+F_4F_5+F_2F_3+F_2F_5+F_2F_3F_4F_5)$ |
| $R_5$ = | $R_{-1}$ | * $(1+F_4F_5+F_2F_3+F_2F_5+F_2F_3F_4F_5)$ | + $R_0$ | * $(-F_3-F_5-F_3F_4F_5-F_1-F_1F_4F_5-F_1F_2F_3-F_1F_2F_5-F_1F_2F_3F_4F_5)$ |

The inverse of $R_0$ mod $R_{-1}$, or 95 mod 357 is the $COEFF_0$ in last round $(-F_3-F_5-F_3F_4F_5-F_1-F_1F_4F_5-F_1F_2F_3-F_1F_2F_5-F_1F_2F_3F_4F_5)$

```
COEFF[3] = 1  ,        COEFF[4] = -F5
    \___________________/____        \*(-F4)
    /                        \+        \
COEFF[2] = COEFF[4]         , COEFF[3] = COEFF[3] -F4*COEFF[4]
    \___________________/____        \*(-F3)
    /                        \+        \
COEFF[1] = COEFF[3]         , COEFF[2] = COEFF[2] -F3*COEFF[3]
    \___________________/____        \*(-F2)
    /                        \+        \
COEFF[0] = COEFF[2]         , COEFF[1] = COEFF[1] -F2*COEFF[2]
```

```
    \__________________/____        \*(-F1)
    /                       \+        \
COEFF[-1] = COEFF[1]         , COEFF[0] = COEFF[0] -F2*COEFF[1]
```